

# 2026 International Mathematical Olympiad

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The 67th International Mathematical Olympiad (IMO) was successfully held in Shanghai, China. As one of the most prestigious and intellectually demanding competitions for secondary school students, the Olympiad brings together the world's top contestants to tackle six extraordinarily challenging problems. This article presents complete solutions to all six problems from the 2026 IMO.

## Day 1

**Problem 1** There are 2026 integers greater than 1 written on a blackboard, repetitions allowed. In one operation, Kong Fuzi chooses two integers  $m > 1$  and  $n > 1$  at different positions on the blackboard, and replaces these two integers with  $\gcd(m, n)$  and  $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$ . As long as such an operation is possible, he continues.

- (1) Prove that no matter how Kong Fuzi chooses, after a finite number of operations, there will be exactly one integer on the blackboard that is greater than 1. Denote this number by  $M$ .
  - (2) Prove that the value of the integer  $M$  is independent of Kong Fuzi's choices.
- $\gcd(x, y)$  denotes the greatest common divisor of positive integers  $x$  and  $y$ , and  $\text{lcm}(x, y)$  denotes the least common multiple of positive integers  $x$  and  $y$ .

**Solution** Let the initial 2026 numbers be  $a_1, \dots, a_{2026}$ . Let the numbers on the blackboard during the process be denoted by  $x_1, \dots, x_{2026}$ . Let

$$P = \prod_{i=1}^{2026} x_i, \quad c = \text{card}\{i | x_i > 1\}.$$

In one operation, we choose  $m, n > 1$ . Let  $g = (m, n)$  and  $l = [m, n]$ . Since  $mn = gl$ , the product of all numbers after the operation becomes

$$P' = P \frac{g(\frac{l}{g})}{mn} = \frac{P}{g}.$$

If  $g = 1$ , then  $m$  and  $n$  are replaced by 1 and  $mn$ , so we have  $P' = P$ ,  $c' = c - 1$ . If  $g > 1$ , then  $P' = \frac{P}{g} < P$ , and among the two newly generated numbers, at least one is  $g > 1$ , so  $c' \leq c$ . Therefore, each operation strictly decreases the positive integer  $P + c$ , so the process cannot go on indefinitely.

Moreover, after each operation, at least one of the new numbers is greater than 1: if  $g > 1$ , then  $g > 1$ ; if  $g = 1$ , then  $\frac{l}{g} = mn > 1$ . Hence, we always have  $c \geq 1$ . When the process stops, if  $c \geq 2$ , we could still choose two numbers greater than 1 to continue the operation, which is a

contradiction. Therefore, when it stops, there is exactly one number greater than 1, which we denote by  $M$ .

For any prime  $p$ , let  $v_p(x)$  denote the exponent of  $p$  in  $x$ . Define  $D_p = \gcd(v_p(x_1), \dots, v_p(x_{2026}))$ . If in one operation  $v_p(m) = \alpha$  and  $v_p(n) = \beta$ , then these two exponents become

$$\min(\alpha, \beta), \quad \max(\alpha, \beta) - \min(\alpha, \beta) = |\alpha - \beta|.$$

Since

$$\gcd(\min(\alpha, \beta), |\alpha - \beta|) = \gcd(\alpha, \beta),$$

$D_p$  remains invariant throughout each operation. At the end of the operation, the numbers on the blackboard are  $M, 1, \dots, 1$ . Therefore,

$$v_p(M) = \gcd(v_p(M), 0, \dots, 0) = \gcd(v_p(a_1), \dots, v_p(a_{2026})).$$

$$M = \prod_{p|a_1 a_2 \dots a_{2026}} p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

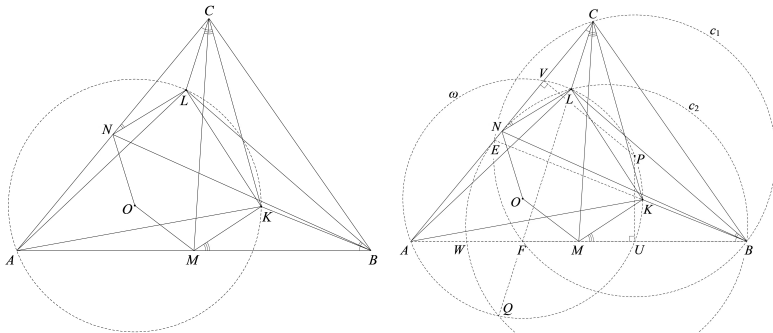
Every prime exponent in  $M$  is uniquely determined by the initial numbers, so  $M$  is independent of Kong Fuzi's choices.  $\square$

**Remark** The greatest common divisor and least common multiple of integers correspond to taking the minimum and taking the maximum respectively in terms of their prime exponent vectors. Therefore, for each prime  $p$ , if the exponents of  $p$  in  $m$  and  $n$  are  $\alpha$  and  $\beta$  respectively, one operation transforms them as

$$(\alpha, \beta) \mapsto (\min(\alpha, \beta), |\alpha - \beta|).$$

This is exactly the Euclidean subtraction process of sorting first and then taking the difference. (1) asks for a non-negative integer that strictly decreases at each step, and (2) asks for a quantity that remains invariant under the aforementioned exponent transformation.

**Problem 2** Let  $M$  and  $N$  be the midpoints of  $AB$  and  $AC$ , respectively. Points  $K$  and  $L$  lie inside  $\triangle BMC$  and  $\triangle BNC$ , respectively, with  $K$  inside  $\triangle LBA$  and  $L$  inside  $\triangle ACK$ . They satisfy  $\angle KBA = \angle ACL$ ,  $\angle LBK = \angle LNC$ , and  $\angle LCK = \angle BMK$ . Let  $O$  be the circumcenter of  $\triangle AKL$ . Prove that  $OM = ON$ .



**Proof** Let  $BK$  intersect  $AC$  at  $E$ , and  $CL$  intersect  $AB$  at  $F$ . Since  $BCEF$  is cyclic, denote this circle by  $c_1$  and its center by  $P$ . Let the projections of  $P$  onto  $AB, AC$  be  $U, V$ , respectively, and let  $W$  be the midpoint of  $AF$ . From  $\angle LBE = \angle LNC$ , we know  $BLNE$  is cyclic. Denote this circle by  $c_2$ . Let  $c_2$  intersect  $CF$  again at  $Q$ . From

$$AW \cdot AB = \frac{1}{2} AB \cdot AF = \frac{1}{2} AC \cdot AE = AN \cdot AE,$$

we deduce that  $W$  lies on  $c_2$ . Since

$$FL \cdot FQ = FB \cdot FW = FU \cdot FA,$$

point  $U$  lies on the circumcircle  $\omega$  of  $\triangle AQL$ . Furthermore, because

$$CA \cdot CV = CN \cdot CE = CL \cdot CQ,$$

point  $V$  also lies on  $\omega$ . Thus,  $AP$  is the diameter of  $\omega$ . Consequently,  $L$  lies on the circle  $\omega$  with diameter  $AP$ . Similarly,  $K$  also lies on  $\omega$ , which means  $\omega$  is also the circumcircle of  $\triangle AKL$ , and hence  $O$  is the midpoint of  $AP$ . Therefore,

$$OM = \frac{1}{2}PB = \frac{1}{2}PC = ON,$$

and the conclusion follows.  $\square$

**Remark** The crux of this problem is establishing a chain of cyclic quadrilaterals and applying the power of a point theorem to show that the circumcircle of  $\triangle AKL$  has  $AP$  as its diameter. By defining intersections  $E, F$  and identifying the center  $P$ , auxiliary points  $U, V$  are systematically proven to lie on the circumcircle  $\omega$ , revealing  $AP$  as its diameter. Once the circumcenter  $O$  is established as the midpoint of  $AP$ , the conclusion is immediate.

**Problem 3** Let  $n$  be a positive integer. Liu Bang and Xiang Yu have a wooden stick of length 1 and wish to divide it. Liu Bang first marks at most  $n$  points on the stick, then Xiang Yu marks at most  $n$  points as well; all marked points are distinct. The stick is then cut at all marked points to obtain several segments. Afterwards, the two players alternately take any unclaimed segment, with Liu Bang moving first. Each player's objective is to maximize the total length of the wooden segments they obtain. For each  $n$ , find the largest real number  $c$  such that Liu Bang can guarantee that his total length is at least  $c$ , regardless of Xiang Yu's actions.

**Proof** First, we show the optimal result. If the final segment lengths are  $x_1 \geq x_2 \geq \dots \geq x_s > 0$ , then taking the current longest segment at each turn is optimal, this can be shown by swapping the behavior of taking a shorter segment with a longer segment. Hence, the total length obtained by Liu Bang is

$$L = x_1 + x_3 + x_5 + \dots = \frac{1 + S}{2}, \quad S = x_1 - x_2 + x_3 - x_4 + \dots \geq 0. \quad (1)$$

**Lemma** Suppose there are initially  $r$  segments of lengths  $a_1, \dots, a_r > 0$ . It is allowed to make at most  $r - 1$  additional cuts. Define

$$d = \min_{I \neq J} \left| \sum_{i \in I} a_i - \sum_{j \in J} a_j \right|,$$

where  $I, J$  range over all different subsets of  $\{1, \dots, r\}$ . Then the minimum achievable value of the alternating difference  $S$  is exactly  $d$ .

Proof of the Lemma: First we prove that it can be done within  $d$ . Take two subsets  $I, J$  whose subset sums differ by  $d$ . Removing their intersection, we can assume  $I \cap J = \emptyset$  and  $\sum_{i \in I} a_i \geq \sum_{j \in J} a_j$ . Arrange the segments in  $I$  end-to-end in a line, and the segments in  $J$  end-to-end in another line. Compare them starting from the same starting point. At existing endpoints on both sides, insert cuts to divide the common portion into several pairs of equal-length small segments. The number of new cuts needed is at most  $|I| + |J| - 1$ . For each segment

not in  $I \cup J$ , cut it in half to obtain a pair of equal segments, requiring  $r - |I| - |J|$  cuts, totaling at most  $r - 1$ . After removing all equal pairs the total length of the remaining segments is at most  $d$ , in the alternating sum sorted by length, each pair of equal segments cancels out, so  $S \leq d$ .

On the other hand, fix how each original segment is cut into pieces and the order of the lengths of all small segments, and move the cut points while maintaining this order. If the small segments on either side of an internal cut point have opposite signs in the alternating sum, move in the direction that decreases  $S$  until a segment length becomes 0 or a pair of opposite-signed equal-length segments appears. If they have the same sign, moving the cut point does not change  $S$ , so we can also move until the above critical situation occurs. Continuously delete zero-length segments and opposite-signed equal pairs. Eventually, no true cut points remain, and the remaining alternating sum is exactly the difference between two subset sums of distinct original segments. Thus, for any cutting method,  $S \geq d$ .

This proves the lemma.  $\square$

For Liu Bang (Guarantee Strategy), Let  $T = 1 + 2 + \dots + 2^n = 2^{n+1} - 1$ . Liu Bang uses  $n$  marked points to cut the stick into  $n + 1$  initial segments with lengths  $\frac{1}{T}, \frac{2}{T}, \frac{4}{T}, \dots, \frac{2^n}{T}$ . Any subset sum of these lengths is  $\frac{k}{T}$  ( $0 \leq k \leq T$ ), and due to the uniqueness of binary representation, they are all distinct. Thus the minimum distance between two distinct subset sums is  $d = \frac{1}{T}$ . Xiang Yu can make at most  $n$  additional cuts. From the lemma, we know that the final result must satisfy  $S \geq \frac{1}{T}$ . Substituting this into (1), Liu Bang can guarantee

$$L \geq \frac{1}{2} \left( 1 + \frac{1}{T} \right) = \frac{2^n}{2^{n+1} - 1}. \quad (2)$$

For Xiang Yu (Restriction Strategy), If Liu Bang cuts out exactly  $n + 1$  initial segments, consider their  $2^{n+1}$  subset sums. All these numbers lie within the interval  $[0, 1]$ . Therefore, when arranged in ascending order, there must be two adjacent numbers whose distance does not exceed  $\frac{1}{2^{n+1}-1}$ .

By the Lemma, Xiang Yu can use at most  $n$  cuts to ensure  $S \leq \frac{1}{2^{n+1}-1}$ . Thus Liu Bang's gain cannot exceed the right-hand side of (2). If Liu Bang only cuts out  $r \leq n$  initial segments, Xiang Yu cuts each initial segment in half, using  $r \leq n$  cuts in total. At this point, all small segments form equal-length pairs, so  $S = 0$ , and Liu Bang can only obtain  $1/2$ , which certainly does not exceed (2).

In summary, the final answer is  $c = \frac{2^n}{2^{n+1} - 1}$ .

Liu Bang's optimal initial cutting method is to divide the wooden stick into  $n + 1$  segments with binary weights  $1 : 2 : 4 : \dots : 2^n$ .  $\square$

**Remark** After cutting, arrange the segment lengths as  $x_1 \geq x_2 \geq \dots$ . When picking alternately, it is optimal to take the current longest segment each time, so Liu Bang obtains the sum of the odd-indexed segments. Thus the problem becomes: Liu Bang first provides at most  $n + 1$  initial segment lengths, and Xiang Yu then cuts them further with at most  $n$  extra cuts. The two players wish to maximize and minimize the final alternating difference  $S$ , respectively. The key observation is that the optimal alternating difference for  $r$  initial segments is controlled by the minimum distance between two subset sums of their lengths. Liu Bang should make all subset sums uniformly distributed the binary weights  $1, 2, 4, \dots, 2^{r-1}$  achieve exactly this.

## Day 2

**Problem 4** Shanyu and Mulan play the following game. Both are given a known angle

$\theta$ , where  $0^\circ < \theta < 180^\circ$ . At the start, Shanyu can arbitrarily form a paper triangle  $T$ . The following steps are then repeated:

1. If the current triangle  $T$  has at least one angle exactly equal to  $\theta$ , the game ends and Mulan wins.
2. Otherwise, Mulan chooses a point  $P$  on the boundary of  $T$  that is not one of the three vertices, and draws a straight line from  $P$  to the opposite vertex on its side, splitting  $T$  into two triangles.
3. Shanyu discards one of the two triangles; the remaining triangle becomes the new  $T$ .

Question: For which real numbers  $\theta$ , regardless of how Shanyu chooses, can Mulan guarantee a win within a finite number of steps?

**Solution** The answer is  $\theta = \frac{180^\circ}{n} \quad (n = 2, 3, \dots)$ .

Suppose Mulan cuts from the vertex of angle  $a$  and divides  $a$  into  $x, y > 0$ , where  $x + y = a$ , the other two angles are  $b, c$ . The three angles of the two sub-triangles are  $(x, b, c + y)$  and  $(y, c, b + x)$ , respectively.

**Lemma** If the current triangle has an angle equal to  $m\theta$ , where  $m$  is a positive integer, then Mulan can guarantee a win within a finite number of steps.

Proof of Lemma: When  $m = 1$ , Mulan has already won. Let  $m \geq 2$ . Mulan cuts from the vertex of this angle, dividing it into  $\theta$  and  $(m - 1)\theta$ . One sub-triangle contains angle  $\theta$ ; the other contains angle  $(m - 1)\theta$ . If Shanyu does not want to immediately keep the former, he can only keep the latter. Repeating this process, the integer  $m$  decreases by 1 each time, so Mulan will win after at most  $m - 1$  additional cuts.

Next we prove the sufficiency. Let  $\theta = \frac{180^\circ}{n}$ .

For  $n = 2$ , i.e.,  $\theta = 90^\circ$ . If the current triangle does not yet have a right angle, let its largest angle be  $a$ , and the other two angles be  $b, c$ , so  $b, c < 90^\circ$ . Let  $x = 90^\circ - b$  and  $y = 90^\circ - c$ . Then  $x, y > 0$  and  $x + y = a$ . Mulan divides angle  $a$  accordingly, the angles of the two sub-triangles at the cut point are  $c + y = 90^\circ$  and  $b + x = 90^\circ$ , respectively. Thus, regardless of which one Shanyu keeps, Mulan wins immediately.

For  $n \geq 3$ : If the current triangle already has an angle equal to  $m\theta$ , then Mulan wins by the lemma. Otherwise, there must currently be an angle  $a > \theta$ , this is because the triangle angle sum is  $n\theta > 3\theta$ , when  $n = 3$ , if none of the three angles is greater than  $\theta$ , they must all be equal to  $\theta$ , contradicting that Mulan has not yet won.

Let the other two angles be  $b, c$ . The open interval  $(b, b + a)$  has length  $a > \theta$ , so it must contain some integer multiple of  $\theta$ , i.e., there exists an integer  $k$  such that  $b < k\theta < b + a$ . Let  $x = k\theta - b$  and  $y = a - x$ . Then  $x, y > 0$ , so angle  $a$  can be divided into  $x, y$ , the two sub-triangles contain angles

$$b + x = k\theta, \quad c + y = a + b + c - k\theta = 180^\circ - k\theta = (n - k)\theta.$$

Also, since  $0 < b < k\theta < b + a < 180^\circ = n\theta$ , we have  $1 \leq k \leq n - 1$ . Regardless of which sub-triangle Shanyu keeps, it contains a positive integer multiple of  $\theta$ , and by the lemma, Mulan can definitely win within a finite number of steps.

Finally we prove the necessity. Suppose  $\theta \neq \frac{180^\circ}{n}$  for every integer  $n \geq 2$ . Shanyu first takes an equilateral triangle. Each of its angles  $60^\circ$  is not a positive integer multiple of  $\theta$ ; otherwise, if  $60^\circ = m\theta$ , then  $180^\circ = 3m\theta$ , contradicting the assumption.

Shanyu maintains the following invariant: none of the three angles of the current triangle is a positive integer multiple of  $\theta$ . Let the current triangle's angles be  $a, b, c$ , and Mulan divides  $a$

into  $x + y = a$ . From (1), the two sub-triangles are  $(x, b, c + y)$  and  $(y, c, b + x)$ . Assume both sub-triangles violate the invariant. Since  $b, c$  were originally not positive integer multiples of  $\theta$ , the first sub-triangle must have either  $x$  or  $c + y$  as a positive integer multiple of  $\theta$ , and the second sub-triangle must have either  $y$  or  $b + x$  as a positive integer multiple of  $\theta$ . Thus, there are only the following four combinations:

1.  $x, y$  are both positive integer multiples of  $\theta$ ,  $a = x + y$  is also;
2.  $x, b + x$  are both positive integer multiples of  $\theta$ ,  $b = (b + x) - x$  is also;
3.  $c + y, y$  are both positive integer multiples of  $\theta$ ,  $c = (c + y) - y$  is also;
4.  $c + y, b + x$  are both positive integer multiples of  $\theta$ ,  $(c + y) + (b + x) = a + b + c = 180^\circ$  is also a positive integer multiple of  $\theta$ .

The first three cases all contradict the current invariant and the fourth case implies  $\theta = \frac{180^\circ}{N}$ , which also contradicts the assumption. Therefore, at least one of the two sub-triangles still satisfies the invariant, and Shanyu can always keep it. In this way the angle  $\theta$  will never appear, so Mulan cannot guarantee a win in a finite number of steps. The necessity is proven.  $\square$

**Remark** The essence of one cut is to divide an angle  $a$  of a triangle into  $x + y = a$ . Mulan must simultaneously control the two branches that Shanyu might keep. The key is to consider the appearance of a positive integer multiple of  $\theta$ , say  $m\theta$  as a winning condition that can be stepwise reduced: cut  $m\theta$  into  $\theta + (m - 1)\theta$ . One branch already has  $\theta$ ; if Shanyu avoids it, he can only keep the branch containing  $(m - 1)\theta$ . Thus, the integer  $m$  decreases successively.

**Problem 5** Let  $\mathbb{R}_{>0}$  denote the set of positive real numbers. Find all functions  $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$  such that for any  $x, y > 0$ ,

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}.$$

**Solution** First we prove the necessity. Take an arbitrary  $t > 0$ . Substitute  $x = f(t)$  and  $y = t$  into the original inequality. Both ends of the inequality chain are equal to  $f(t)$ , so the middle term must also be equal to  $f(t)$ , thus  $f(f(t)) = 2f(t) - t$ . Since  $f(f(t)) > 0$ , this equation also implies  $f(t) > t/2$ . Fix  $t > 0$  and let  $a = f(t)$ . Take an arbitrary  $h > 0$ . Substitute  $x = a$  and  $y = t + h$  into the original inequality. Since  $f(a) = 2a - t$ , so we have

$$\sqrt{\frac{a^2 + f(t+h)^2}{2}} \geq \frac{f(a) + t + h}{2} = a + \frac{h}{2} \geq \sqrt{af(t+h)}.$$

By squaring the two inequalities separately, we obtain

$$\sqrt{a^2 + 2ah + \frac{h^2}{2}} \leq f(t+h) \leq a + h + \frac{h^2}{4a}.$$

Note that

$$(a + h) - \sqrt{a^2 + 2ah + \frac{h^2}{2}} = \frac{h^2/2}{a + h + \sqrt{a^2 + 2ah + h^2/2}} \leq \frac{h^2}{2a}.$$

Therefore,

$$|f(t+h) - f(t) - h| \leq \frac{h^2}{2f(t)}.$$

Now take arbitrary  $0 < x < y$ , and divide  $[x, y]$  into  $N$  segments. Let  $h = (y - x)/N$ ,  $t_k = x + kh$  ( $k = 0, 1, \dots, N$ ). From  $f(t_k) > t_k/2 \geq x/2$  we get

$$|f(t_{k+1}) - f(t_k) - h| \leq \frac{h^2}{2f(t_k)} < \frac{h^2}{x}.$$

Adding these inequalities for  $k = 0, 1, \dots, N - 1$ , and using the triangle inequality, we get

$$|f(y) - f(x) - (y - x)| \leq \frac{Nh^2}{x} = \frac{(y - x)^2}{Nx}.$$

This inequality holds for every positive integer  $N$ . Letting  $N$  tend to infinity, the right-hand side tends to 0, so  $f(y) - f(x) = y - x$ . Therefore,  $f(x) - x$  is a constant on  $\mathbb{R}_{>0}$ , that is, there exists a real constant  $c$  such that  $f(x) = x + c$  ( $x > 0$ ). Since  $f(x) > 0$  holds for all  $x > 0$ , we must have  $c \geq 0$ .

Finally, verify the sufficiency. Take any  $c \geq 0$  and let  $f(x) = x + c$ . Then  $f(x) + y = x + f(y)$ . By the inequality that Root Mean Square is not less than Arithmetic Mean, and Arithmetic Mean is not less than Geometric Mean, we obtain

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{x + f(y)}{2} = \frac{f(x) + y}{2} \geq \sqrt{xf(y)}.$$

Therefore, the complete set of solutions is  $\boxed{f(x) = x + c \quad (c \geq 0)}$ . □

**Remark** Let  $x = f(y)$ , then both sides are simultaneously equal to  $f(y)$ , forcing the middle term to also be equal to  $f(y)$ , leading to the key iterative equation  $f(f(y)) = 2f(y) - y$ . Next, fix  $t > 0$ , let  $a = f(t)$ , and substitute  $x = a$ ,  $y = t + h$ . The crucial iterative equation turns the middle term accurately into  $a + h/2$ . Finally, divide any interval  $[x, y]$  into  $N$  equal subintervals, the error for each segment is a quadratic small term, so the total error over all  $N$  segments is only  $O(1/N)$ . By allowing  $N$  to grow arbitrarily large, we obtain  $f(y) - f(x) = y - x$ .

**Problem 6** Let  $a_1, a_2, a_3, \dots$  be an infinite sequence of positive integers greater than 1. It is known that for every positive integer  $n$ ,  $a_{n+1}$  is the smallest positive integer satisfying the following conditions:  $a_{n+1} > a_n$ , and for each  $i = 1, 2, \dots, n$ , we have  $\gcd(a_{n+1}, a_i) > 1$ . Prove that there exist positive integers  $T, L$  such that for every positive integer  $n$ , we have  $a_{n+T} = a_n + L$ .

**Solution** Denote  $\mathcal{A} = \{a_1, a_2, \dots\}$ ,  $P(m) = \{\text{all prime divisors of } m\}$ . From the recursive condition, any two terms in the sequence are not coprime, i.e.

$$i \neq j \implies P(a_i) \cap P(a_j) \neq \emptyset. \quad (3)$$

If an integer  $m \geq a_1$  does not belong to  $\mathcal{A}$ , let  $k$  be such that  $a_k < m < a_{k+1}$ . By the minimality of  $a_{k+1}$ , there must exist some  $i \leq k$  satisfying  $\gcd(m, a_i) = 1$ . Hence,

$$m \notin \mathcal{A} \implies \exists a_i < m, \quad P(m) \cap P(a_i) = \emptyset. \quad (4)$$

This yields the important property: If  $x \in \mathcal{A}$ ,  $m \geq a_1$ , and  $P(x) \subseteq P(m)$ , then  $m \in \mathcal{A}$ . Otherwise, there exists  $a_i$  coprime to  $m$ , and hence coprime to  $x$ , contradicting (3). Now sequentially examine  $a_1, a_2, \dots$ , first choose  $a_1$ , if no previously chosen critical term  $b$  satisfies  $P(b) \subseteq P(a_n)$ , then choose  $a_n$ . Clearly, every term  $a_n$  in the sequence contains all the prime factors of some critical term. We now prove that there are only finitely many critical terms. Let  $Q = P(a_1)$ ,  $q_0 = \max Q$ ,  $C = a_1 q_0$ . Suppose a critical term  $x > C$  contains a prime factor  $p$  that has never appeared in any previous critical terms. Since  $a_1$  is a previous critical term,  $p \notin Q$ . Also, since  $\gcd(x, a_1) > 1$ , we can choose  $q \in P(x) \cap Q$  ( $q \neq p$ ). Let

$$S = P(x) \setminus \{p\}, \quad r = \prod_{\ell \in S} \ell.$$

For any previous critical term  $b$ , by (3) we have  $P(b) \cap P(x) \neq \emptyset$ . Since  $p \notin P(b)$ , we get  $P(b) \cap S \neq \emptyset$ . Every previous term in the sequence contains all the prime factors of some

previous critical term; hence, the set of its prime factors is  $S$ . Any integer with prime factors in  $S$  is coprime to all the terms in the sequence before  $x$ . If  $r \geq a_1$ , take  $y = r$ . Since  $r \mid x$ , we have  $a_1 \leq y = r < x$ . If  $r < a_1$ , take the smallest positive integer  $t$  such that  $y = rq^t \geq a_1$ . Since  $rq^{t-1} < a_1$ , we have

$$a_1 \leq y = rq^t < qa_1 \leq q_0a_1 = C < x.$$

Thus, in both cases, we have  $a_1 \leq y < x$ ,  $P(y) = S$ . Since  $y$  is coprime to all previous terms before  $x$ , and by (4), we know  $y \in \mathcal{A}$ . Hence,  $y$  contains all the prime factors of some previous critical term  $b$ , so

$$P(b) \subseteq P(y) = S \subseteq P(x),$$

which contradicts the fact that  $x$  is a critical term. Therefore, no critical term greater than  $C$  can introduce a new prime factor. There are only finitely many critical terms not exceeding  $C$ , let the union of their prime factors be the finite set  $K$ . By the conclusion derived above, all subsequent critical terms  $x$  satisfy  $P(x) \subseteq K$ . Since  $K$  has only finitely many subsets, and the prime factor sets of distinct critical terms must be distinct, there are only finitely many critical terms in total. Let all critical terms be  $b_1, b_2, \dots, b_s$ , and set

$$d_j = \prod_{p \in P(b_j)} p = \text{rad}(b_j) \quad (1 \leq j \leq s).$$

From the definition of critical terms and property (3), for any  $m \geq a_1$ ,

$$m \in \mathcal{A} \iff \exists j \in \{1, \dots, s\}, \quad d_j \mid m.$$

Take  $L = \text{lcm}(d_1, \dots, d_s)$ . Since each  $d_j \mid L$ , we get

$$m \in \mathcal{A} \iff m + L \in \mathcal{A} \quad (m \geq a_1).$$

Finally, let  $T = \text{card}(\mathcal{A} \cap [a_1, a_1 + L))$ . Note that the mapping  $m \mapsto m + L$  provides an order-preserving bijection from  $\mathcal{A}$  onto  $\mathcal{A} \cap [a_1 + L, \infty)$ . Therefore, after removing the first  $T$  terms, the remaining terms are exactly  $a_1 + L, a_2 + L, \dots$ . Thus, for any positive integer  $n$ , we have  $a_{n+T} = a_n + L$ .  $\square$

**Remark** Visualize the generation process of the sequence as scanning integers one by one starting from  $a_1$ . The core of the proof is to strengthen the condition any two terms are not coprime. In fact, any two terms in the sequence must share a common prime factor not exceeding  $A$ . Suppose there exists a counterexample  $u < v$ , and choose the one with the smallest larger term  $v$ . By removing all large prime factors of  $u$  and raising a suitable small prime factor of  $u$  to an appropriate power, we can construct an integer  $x$  in  $[A, u)$  that only retains the small prime factor characteristics of  $u$ . Since  $x$  is coprime to  $v$ , it must have been skipped by the sequence, hence there exists an earlier term  $w < x$  that is coprime to  $x$ , turning  $(w, u)$  into a smaller counterexample and yielding a descent contradiction.

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